MIL-C-29135A
14 August 1980
SUPERSEDING
MIL-C-29135
10 September 1974

MILITARY SPECIFICATION

CLOTH, KNITTED, NYLON, SIMPLEX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for a simplex knitted nylon cloth.

1.2 Classification The cloth shall be of one type only, and in the following classes as specified (see 6.2):

Class 1 - White 3059
Class 2 - Black 3231

2. APPLICABLE DOCUMENTS

2.1 Issue of document The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, or by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1491 - Glossary of Knitting Imperfections

(Copies of specifications, standards, drawings, and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580).

3. REQUIREMENTS

3.1 Standard sample The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.2 First article When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material

*3.3.1 Yarn The yarn used shall be of a multifilament 40 to 70 denier, semi-dull or dull, nylon (see 4.5). All yarn utilized for any one contract shall be of the same denier, luster and number of filaments.

3.4 Color

*3.4.1 Class 1 The class 1 cloth shall be white 3059 to match the standard sample (see 6.4). The cloth shall be fully bleached and may be supplemented with a fluorescent optical brightener to the blue region. The bleached cloth shall not discolor to a degree greater than that shown by the standard sample when tested as specified in 4.5. When a standard sample is not available, the discoloration to light shall be no lower than a rating of "fair".

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3.4.2 Class 2 The class 2 cloth shall be black 3231, and shall match the standard sample (see 6.4). The dyed and finished cloth shall show fastness to light, laundering, perspiration, and crocking equal to or better than the standard sample. When a standard sample is not available, the finished cloth shall show a minimum of "good" fastness to light, laundering, and perspiration; and shall show a Munsell Value for crocking not lower than 8.5. Testing shall be as specified in 4.5.

3.4.3 Matching The color of the finished cloth shall match the standard sample under artificial daylight having a color temperature of 7500° Kelvin, and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

*3.5 Physical requirements The finished cloth shall be of a simplex plain jersey stitch pattern (see 6.5), conforming to the requirements specified in Table I.

Table I - Physical requirements

Characteristic	Requirements	
	Minimum	Maximum
Weight, ounces per square yard	5.5	8.0
Wales per inch	52	-
Courses per inch	40	-
Bursting strength, pounds	100	-
Stretch percent, length (wale) direction	-	30.0
Stretch percent, width (course) direction	45.0	75.0

3.5.1 Width The width of the finished cloth shall be as specified (see 6.2)

*3.6 Finish The finished cloth shall have a smooth sueded finish on both sides of the material, and shall be fully heat set to meet the requirements specified in 3.6.1.

3.6.1 Dimensional stability The finished cloth shall not shrink more than 10% in the length (wale) direction nor more than 10% in the width (course) direction when tested as specified in 4.5.

3.7 Non-fibrous materials The total chloroform-soluble and water-soluble material content of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.5.

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3.8 pH The pH value of the finished cloth shall be not lower than 5.0 nor higher than 8.5 when tested as specified in 4.5.

3.9 Length and put-up Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 35 yards. Each length shall be put-up in full width rolls as specified in PPP-P-1133.

3.10 Fiber identification Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act (see 2.2).

3.11 Workmanship The finished cloth shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

*4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection When required, the first article of the finished cloth submitted in accordance with 3.2, shall be visually inspected for shade, finish, and appearance, and shall be tested for chemical and physical properties in accordance with 4.4.

4.4 Quality conformance inspection Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

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4.4.1 Component and material inspection In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise amended, excluded, modified, or qualified in this specification or applicable purchase document.

4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.3.

4.4.2.1 Yard-by-yard examination The required yardage of each roll shall be examined and visual defects as defined in MIL-STD-1491 shall be classified as listed below. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each wale-wise (length-wise) yard or fraction thereof in which it occurs. The sample unit shall be one linear yard. The sample size shall be in accordance with inspection level II of MIL-STD-105. The acceptable quality levels shall be 1.5 major defects and 6.5 total defects (major and minor combined) per 100 units (yards). The lot size shall be expressed in units of one linear yard each. The number of rolls selected shall be in accordance with Table II. An approximate equal number of yards shall be examined from each roll in the sample.

Defects	Classification Major	Minor
Hole, cut, tear, float, run or dropped stitch	X	
Tuck stitch, pulled stitch, or needle streak		X
Any visible mend	X	
Abrasion mark resulting in a weak area	X	
Thin or uneven area	X	
Coarse or fine yarn 1/		X
Tight twist yarn 1/		X
Crease embedded in cloth	X	
Any knot protruding above surface of cloth		X
Barre mark 1/	X	
Spot or stain	X	
Kinks or untrimmed ends		X
Cockled (ridgy) cloth	X	
Slubby yarn more than twice the size of the normal diameter of the yarn		X
Knitted in waste 1/		X

1/ Clearly visible at normal inspection distance (approximately three feet).

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4.4.2.2 Overall examination Each defect listed below shall be counted not more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the number of defects acceptable shall be as shown in Table II.

Defects

Overall width less than specified.
Objectionable odor.
Overall uncleanness.
Uneven knitting throughout the roll.
Off shade, shaded end to end, side to side, side to center, or throughout the roll.
Poor dye penetration, mottled, streaky or cloudy.
Not labeled in accordance with the Textile Fiber Products Identification Act.
Edges frayed, ravelled, curled, rolled, folded, scalloped, or uneven continuously or intermittently throughout the roll.
Sueded finish omitted on one side of the material.

Table II

Lot size (yards)	Sample size (rolls)	Maximum number of defects acceptable in sample
Up to 1200	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

4.4.2.3 Examination for length

4.4.2.3.1 Individual rolls The rolls shall be examined for gross length. Any gross length found to be less than the minimum specified or more than 2 yards less than the gross length marked on the ticket shall be considered a defects with respect to length. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptable number shall be as shown in Table II.

4.4.2.3.2 Total yardage in sample The lot shall be unacceptable if the total of the actual gross lengths of rolls in the sample is less than the total of the gross lengths marked on the roll tickets.

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4.4.3 Examination of packaging requirements An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing, and marking comply with Section 5 requirements.

4.5 Testing of the end item The methods of testing specified in FED-STD-191 whenever applicable, and as listed in Table IV shall be followed. The physical and chemical values specified in Section 3, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3 1/2 continuous yards full width of the finished cloth. The lot size shall be expressed in units of 1 yard each. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual value utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with Table III.

Table III - Sampling for testing

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table IV - Test methods

Characteristic	Requirement paragraph	Test method
Nylon identification	3.3.1	1530 1/
Yarn:		
Denier	3.3.1	4021 1/
Multifilament	3.3.1	Visual 1/
Luster	3.3.1	Visual 1/
Discoloration to (class 1)		
Light	3.4.1	5660 2/
Colorfastness to (class 2)		
Light	3.4.2	5660
Laundering	3.4.2	5614
Perspiration	3.4.2	5680
Crocking	3.4.2	5651
Weight	3.5	5041
Wales per inch	3.5	5070
Courses per inch	3.5	5070

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Table IV - Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Bursting strength	3.5	5120
Stretch:		
Length (wale) direction	3.5	4.5.1
Width (course) direction	3.5	4.5.1
Type of knit	3.5	Visual 1/
Dimensional stability:		
Length	3.6.1	5552
Width	3.6.1	5552
Nonfibrous material	3.7	2611
pH	3.8	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Method 5660 shall be used except that exposure of the specimen and standard sample shall be for 20 standard fading hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "pass". Any specimen showing discoloration greater than that of the standard sample shall be rated "fail". One determination shall be made for each sample unit and the results reported as "pass" or "fail".

4.5.1 Test for stretch

4.5.1.1 Test specimen The test specimen shall be a rectangle 4 inches by not less than 5 inches. The long dimension shall be parallel to the length (wales) of the cloth for length tests and parallel to the width (courses) for width tests. The test specimen shall be taken no nearer the selvage than 1/10 the width of the cloth. Two parallel lines, 3 inches apart, shall be drawn at right angles to the long dimension of the specimen. In addition, a line shall be drawn parallel to the long dimension of the specimen and 1 inch in from either edge for properly aligning the specimen in the jaws.

4.5.1.2 Test apparatus The test device (Figure 1) shall be an apparatus wherein the specimen is held between two clamps. The design of the upper clamp shall be such that one gripping surface or jaw shall be an integral part of the rigid frame of the clamp. The face of the movable jaw shall measure 1 inch by 1 inch and the face of the fixed jaw shall measure 1 inch by 1 1/2 or more inches with the long dimension perpendicular to the direction of application of the load. The lower clamp shall have similar jaws and when weight is attached, shall weigh a total of 6 pounds. The jaws shall have smooth gripping surfaces, sufficiently flat and parallel to prevent slipping of the specimen during the test.

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4.5.1.3 Procedure The specimen shall be placed symmetrically in the clamps of the apparatus with the long dimension parallel and the short dimension at right angles to the direction of the application of the load. Using the specified guidelines, the specimen shall be aligned with one outside edge of the front jaws of each clamp. The proper weight shall be added to insure a total of 6 pounds (including the weight of the lower jaws). The load shall be applied for 15 seconds and the distance between jaws measured by a pair of dividers or other suitable instrument. The elongation at the required load shall be expressed as the percent increase in the length of the specimen between the jaws. The percent stretch shall be calculated as follows:

$$\text{Percent strength} = \frac{\text{Stretch dimension} - \text{Original 3-inch dimension}}{\text{Original dimension (3 inches)}} \times 100$$

4.5.1.4 Report. One specimen each from the length (wales) and width (courses) direction shall be tested from each sample unit and the results obtained reported separately to the nearest percent.

5. PACKAGING

5.1 Preservation-packaging Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A and C The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking In addition to any special marking required by contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

6. NOTES

6.1 Intended use The cloth covered by this specification is intended for use in the fabrication of gloves.

6.2 Ordering data Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Class (color) required (see 1.2)
c. When first article is required (see 3.2).
d. Width of cloth required (see 3.5.1).
e. Minimum length of cloth if other than specified (see 3.9).
f. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 Contract data requirements Data conforming to Department of Defense Data Item Description DI-R-4803, DI-R-4805, DI-T-4901, DI-T-4902, DI-T-4903 and DI-T-4904 will usually be required for delivery in connection with this specification for Marine Corps procurements. For other than Marine Corps procurement, the coginizant procuring activity will provide comparable Data Item Descriptions. When so required, such data will be specified for delivery on a DD-1423 included in the contract.

6.4 Standard sample For access to standard samples, address the procuring office issuing the invitation for bids.

6.5 Knitting machine The cloth can be made on a 30 or 32 gauge, two-needle bar, warp (simplex) knitting machine.

Custodians:
Navy - NU
Army - GL
Air Force - 11

Review activities:
Navy - MC
Army - MD
Air Force - 82

User activity:
Navy - CG

Preparing activity:
Navy - NU

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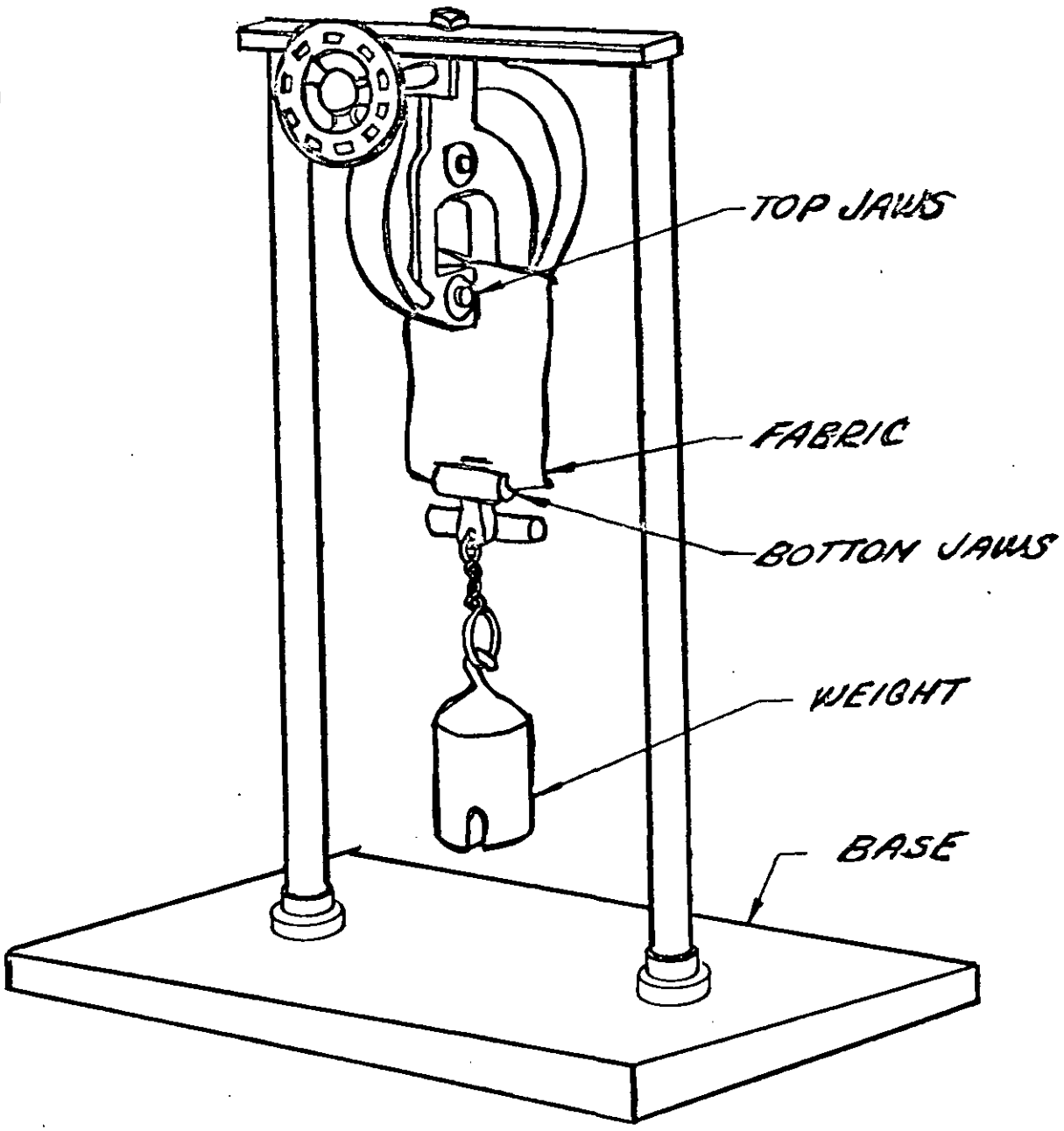


FIGURE 1. FABRIC STRETCH APPARATUS

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-29135A CLOTH, KNITTED, NYLON, SIMPLEX

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

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DATE

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